

Spectral Interpretability of a Stellar Transformer

Project Summary — Sim-to-Real G/K Classification with TransformerPayne

Course AS4501 (Astroinformatics) • sim→real branch • Emulator: TransformerPayne • Range 4000–5000 Å

1. Overview and Objectives

Stars are classified by spectral type, a temperature sequence imprinted as absorption lines in the spectrum. This branch runs a **sim-to-real** experiment with the **TransformerPayne** emulator: generate synthetic spectra, train a Random Forest classifier, apply it to real DESI spectra, and interpret which spectral lines drive the decision via feature importance. The full project also targets a second emulator (The Payne), SHAP, an attention-head analysis, and an extension to more classes; this document covers the implemented sim-to-real pipeline.

Current scope. The team focuses on the **G and K** classes (F is inside range but omitted for time) over **4000–5000 Å** — the valid range of the TransformerPayne paper and the range in which the real DESI spectra were downloaded, so simulated and real data are directly comparable.

2. Background and Theory

2.1 Spectral classification (G vs K)

The spectral type follows effective temperature. Within 4000–5000 Å, the **Balmer lines** (H-delta 4103, H-gamma 4341, H-beta 4863 Å) are stronger in the hotter **G** stars, whereas **metal lines** and the **CH G-band** (~4305 Å) strengthen toward the cooler **K** stars. G and K are **neighboring types**, so separating them is a genuine (non-trivial) classification task.

2.2 TransformerPayne, Random Forest, DESI

TransformerPayne (Rozanski, Ting & Jablonska 2023) is an attention-based emulator mapping stellar parameters to a normalised spectrum; it is trained on **synthetic LTE spectra** at high resolution, and its released weights are valid for G/K dwarfs (Teff ~4000–6000 K) over ~4000–5000 Å. A **Random Forest** (RF) uses each wavelength pixel as a feature; its **feature importance** reveals which wavelengths drive the classification. **DESI** provides the real stellar spectra; cross-matching DESI with Gaia and the DESI-MWS value-added catalog yields real Teff labels (used to build G/K ground truth).

3. Methodology and Key Decisions

- **Common grid:** 4000–5000 Å, 1000 points (~1 Å/pixel), identical for simulated and real spectra (required for the RF to transfer).
- **Physical lines (in-range):** H-delta, Ca I 4227, CH G-band, H-gamma, H-beta.
- **Balanced dataset:** equal numbers of G and K, with unified Gaussian noise in the continuum-normalised space (avoids per-class leakage).
- **Identical continuum normalization** (high-percentile filter) applied to sim and real, so the sim-trained model transfers.
- **Random Forest** trained with GridSearchCV + 5-fold stratified cross-validation.
- **Real DESI pipeline:** SPARCL download / CSV loader, mask bad pixels, continuum-normalize, rest-frame shift, resample to the common grid.
- **LSF broadening** (fixed and wavelength-dependent $R(\lambda)$) to bring the sharp simulated lines to DESI resolution (domain-shift mitigation).
- **Real-label evaluation scaffold:** `teff_to_class + evaluate_on_labeled`, ready for a true accuracy on labelled DESI G/K.

4. Results

4.1 Classification on simulated data.

On the held-out test set the RF reaches **accuracy ≈ 0.958** : **G is classified perfectly and only 2 K spectra are confused with G**. Unlike the trivial 100 % obtained earlier with F/G/K (widely separated temperatures), this is a **meaningful result** on the adjacent G/K pair. The within-class scatter is small (thin $\pm 1\sigma$ band), i.e. the classes are internally homogeneous.

4.2 Interpretability (feature importance).

The RF importance concentrates on **real spectral lines** within 4000–5000 Å: the Balmer series (H-delta, H-gamma, H-beta, stronger in G) plus a forest of metal lines (stronger in K). The model relies on genuine, physically meaningful features — the interpretability goal of the project.

4.3 Sim-to-real transfer.

The sim-trained RF is applied to real DESI spectra. A **domain shift** remains — simulated lines are sharper than the instrumentally broadened, noisier real lines (the emulator is trained on synthetic LTE physics) — and is mitigated by the wavelength-dependent LSF broadening. A **true labelled accuracy** on real DESI G/K will be obtained by feeding the team's balanced, labelled G/K sample (DESI x Gaia x MWS) into `evaluate_on_labeled`.

5. Discussion, Limitations and Status

- **0.958 vs the old 1.000**: moving from F/G/K to the neighboring G/K pair turns a trivial task into a meaningful one; the score reflects real physics (feature importance on Balmer + metal lines), not just a temperature jump.
- **Validity**: restricting to 4000–5000 Å and to G/K keeps everything inside TransformerPayne's valid range, removing the earlier extrapolation concern.
- **Domain shift is fundamental**: the emulator is synthetic-only (LTE), so broadening helps but does not fully close the gap.
- **Open**: real labelled accuracy (labels pending), balanced real set (limited by the scarce G stars), and the large-scale run (equal real and synthetic counts).

Requirement (objective)	Status
1 — synthetic + classifier (TransformerPayne, G/K)	Done
1 — apply to real DESI (sim-to-real)	Done (labelled accuracy pending)
1 — second emulator (The Payne)	Team / pending
2 — SHAP + compare emulators	Pending
3 — attention-head analysis	Team / pending
4 — extend to other classes	Pending

Reference: Rozanski, Ting & Jablonska (2023), *Toward a Spectral Foundation Model*, arXiv:2306.15703. AI assistance was used for code formatting and bilingual text; the scientific approach, decisions and results are the author's own and were reviewed by the author.